Analytic Epidemiology - Study Designs I

APHS 20203 • Epidemiology & Biostatistics

Epidemiological Study: Basic Question
Are exposure and outcome associated?
More specifically:

- Do people with different exposures experience different health outcomes?
- Is the observed association likely to be causal, or could it reflect other explanations?

Classification of Study Designs

Two Categories of Analytic Studies

Observational design

Investigator does not assign exposure

Observes existing differences in exposure and outcome

Primary goal: identify associations

Examples include ecologic, cohort, and case-control studies

Experimental design

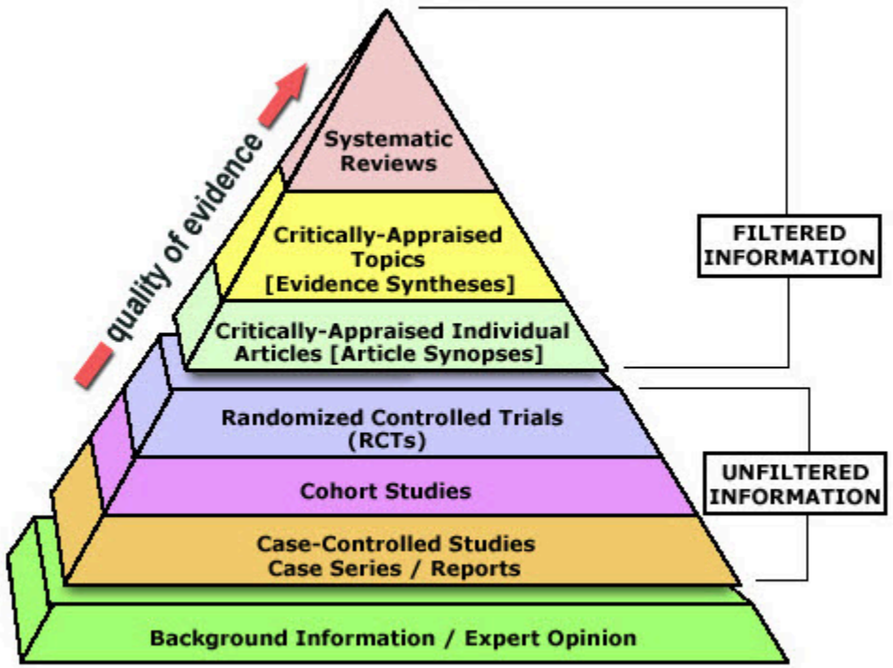
Investigator assigns exposure or intervention

Often uses randomization

Primary goal: evaluate causal effects

Examples include randomized controlled trials (RCTs)

Levels of Evidence in Epidemiologic Research



Generate Hypotheses
Establish Causality

- Study designs vary in how strongly they support causal inference
- Lower levels are often used to generate hypotheses
- Higher levels are better suited to test causal effects
- Moving up the pyramid generally means:
  - Greater control over alternative explanations
  - Stronger ability to evaluate causality

Important note:
Higher evidence does not mean “always correct,” and lower evidence does not mean “wrong.”

Epidemiologic Inference: From Association to Conclusion

Analytic epidemiology asks whether an observed association reflects a causal relationship.

- Step 1: Identify an association: Do exposure and outcome occur together more often than expected?
- Step 2: Evaluate validity:
  - Internal validity: Is the association likely to be real within this study?
  - External validity: Can the findings be generalized to other populations?
- Step 3: Consider alternative explanations
  - Could the association reflect chance?
  - Could it reflect systematic differences in how groups are compared or observed?
  - Could it reflect other factors related to both exposure and outcome?

Key takeaway: Study design and validity assessment determine how confidently we can move from association to inference.

Ecologic Studies: Limits of Causal Inference

- Ecologic Studies: Group-level analysis
- Unit of analysis: groups (not individuals)
- Exposure & outcome: measured at the population level
- Common use: explore population patterns; generate hypotheses
- Key caution: group-level associations ≠ individual-level conclusions
- Take-home message: Ecologic studies are useful for population questions, but cannot establish individual-level causality.

Ecologic Fallacy

Definition: Incorrectly applying group-level findings to individuals

Occurs when associations observed in populations are assumed to hold at the individual level

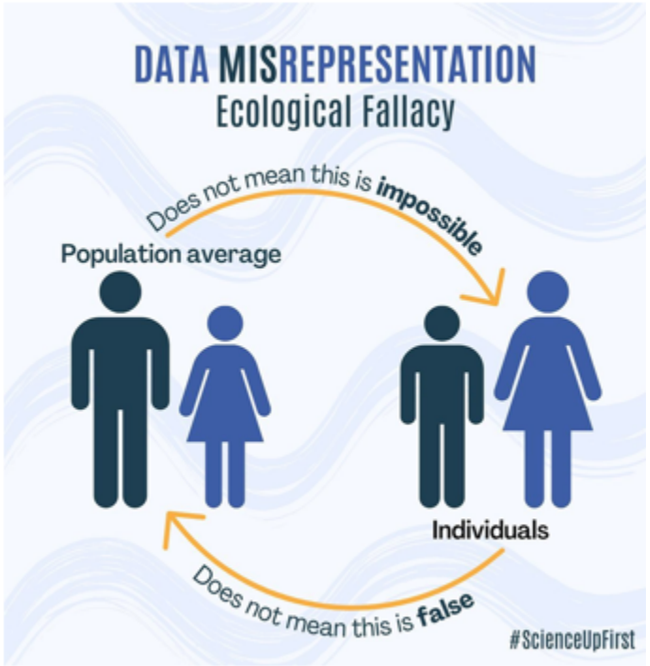
Example

Observation: Cities with higher income have higher CHD rates

Incorrect inference: Wealthier individuals have higher CHD risk

Reality: Within cities, CHD risk may be higher among lower-income individuals

Take-home message: Group-level data cannot be used to draw individual-level conclusions.



https://www.scienceupfirst.com/project/data-misrepresentation-ecological-fallacy/

Ecologic Study: Strengths & Limitations

Strengths	Limitations
Uses existing population-level data	Ecologic fallacy
Efficient: fast and low cost	Crude exposure measurement
Useful when individual-level data are unavailable	Limited control of confounding

Discussion: Ecologic Study Example

- Study
  - Sasaki et al. (1993): dietary fat intake & breast cancer mortality
  - Unit of analysis: countries (30 countries total)
- Key finding
  - Higher animal fat intake was associated with higher breast cancer mortality
  - Fish fat intake was associated with lower breast cancer mortality
- Discussion prompts
  - What makes this study feasible and efficient?
  - Why can these results not be interpreted at the individual level?
  - What other population-level factors could explain the association?

Sasaki S, Horacsek M, Kesteloot H. An ecological study of the relationship between dietary fat intake and breast cancer mortality. Prev Med. 1993 Mar;22(2):187-202. doi: 10.1006/pmed.1993.1016. PMID: 8483858.

Experimental Studies

- Key feature: investigator assigns exposure or intervention
  - Assigning exposure allows stronger conclusions about cause and effect.
- Comparison: intervention group vs control group
- Goal: evaluate the effect of an intervention
- Two broad types
  - True experimental studies: include randomization
  - Quasi-experimental studies: no randomization, but intervention is assigned
- Why they matter
  - Stronger control of alternative explanations
  - Better support for causal inference than observational designs

Intervention Studies

- Key Features
  - A type of experimental study
  - Introduce a planned action to assess its impact
  - Conducted in real-world settings
- Common examples
  - Medical interventions (e.g., vaccines, treatments)
  - Behavioral programs (e.g., smoking cessation)
  - Policy or community interventions (e.g., taxation, education)
- Take-home message: Intervention studies evaluate whether deliberate actions change health outcomes in practice.

Experimental vs Intervention Studies

- Experimental studies
  - Broad category where the investigator manipulates a condition
  - Often conducted in controlled settings
  - Example: laboratory study testing how temperature affects a biological process
- Intervention studies
  - A type of experimental study focused on planned actions to improve health
  - Conducted in real-world populations
  - Example: introducing a vaccination program or a smoking cessation policy
- Key point: All intervention studies are experimental, but not all experimental studies are health interventions.

Exercise: Experimental vs Intervention Studies

- Exercise: For each example below, decide whether it is an Experimental study or an Intervention study.
- Researchers expose bacteria to different temperatures in a lab and measure growth rate.
- A city introduces a new bike-lane policy to encourage physical activity and reduce injuries.
- Participants are randomly assigned to receive a new medication or a placebo to evaluate treatment effectiveness.
- Psychologists test whether blue or red lighting affects concentration in a controlled setting.
- Hint
  - Which examples involve a planned action to improve health?
  - Which examples involve manipulation without a health intervention?

Clinical Trials: Medical Intervention Studies

- Clinical Trials
  - A type of intervention study conducted in humans
  - Participants receive a medical intervention under controlled conditions
  - Designed to evaluate safety and effectiveness
- Common purposes
  - Prevention (prophylactic trials, e.g., vaccines)
  - Treatment (therapeutic trials, e.g., new drugs or procedures)
- Key takeaway: Clinical trials provide strong evidence for whether medical interventions work and are safe.

Randomized Controlled Trials (RCTs)

- Randomized Controlled Trials (RCTs)
  - A type of clinical trial
  - Participants are randomly assigned to intervention or control groups
  - Groups are comparable at baseline by design
  - Outcomes are measured consistently across groups
- Why randomization matters
  - Reduces systematic differences between groups
  - Strengthens causal inference
- Take-home message: Randomization helps isolate the effect of the intervention, making RCTs the strongest design for evaluating causality.

RCT: Randomization

Randomization

Assigns participants to groups by chance

Balances known and unknown factors across groups

Occurs before outcomes are measured

What randomization does

Reduces systematic differences between groups

Minimizes alternative explanations for observed effects

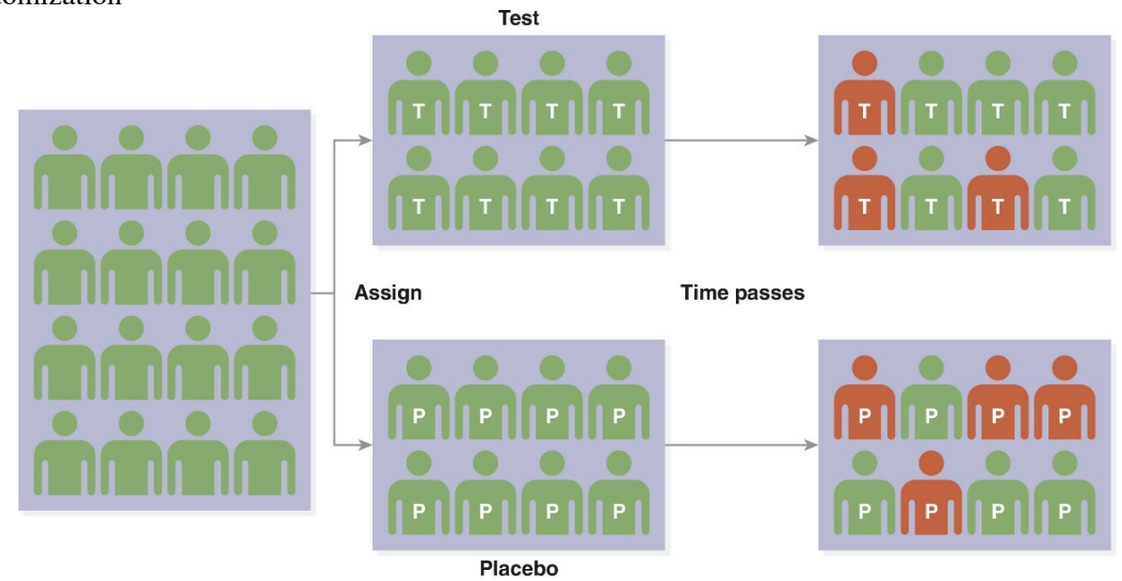


Figure 7.5 Diagram of a randomized controlled trial

Indication that red means occurrence of the outcome.

Take-home message

Randomization improves internal validity by making groups comparable at baseline.

Discussion: Randomized Controlled Trial (RCT) Example

- Study overview
  - Large randomized controlled trial of a low-fat dietary pattern
  - Postmenopausal women assigned to intervention or usual diet
  - Long-term follow-up
- Discussion prompts
  - Why is an RCT an appropriate design for this question?
  - What features of this study strengthen causal inference?
  - What challenges might arise in long-term behavioral interventions?
- Key takeaway: RCTs strengthen causal conclusions, but feasibility, adherence, and follow-up matter.

Prentice RL, Caan B, Chlebowski RT, et al. Low-Fat Dietary Pattern and Risk of Invasive Breast Cancer: The Women's Health Initiative Randomized Controlled Dietary Modification Trial. JAMA. 2006;295(6):629–642. doi:10.1001/jama.295.6.629

Discussion: Comparing Study Designs

- Comparing Two Studies: Think back to the two studies discussed today:
  - Ecologic study: dietary fat intake and breast cancer mortality
  - Randomized controlled trial: low-fat dietary intervention
- Discussion questions
  - What is the unit of analysis in each study?
  - Which study provides stronger evidence for causality, and why?
  - What kinds of conclusions are appropriate for each design?

RCTs: Strengths & Limitations

Strengths	Limitations
Randomization improves group comparability	Often expensive and time-consuming
Strong internal validity	May have limited generalizability to real-world populations
Best design for evaluating causal effects of interventions	Not always ethical or feasible

Project Prep 2

Revisit: Study Design & Causal Inference

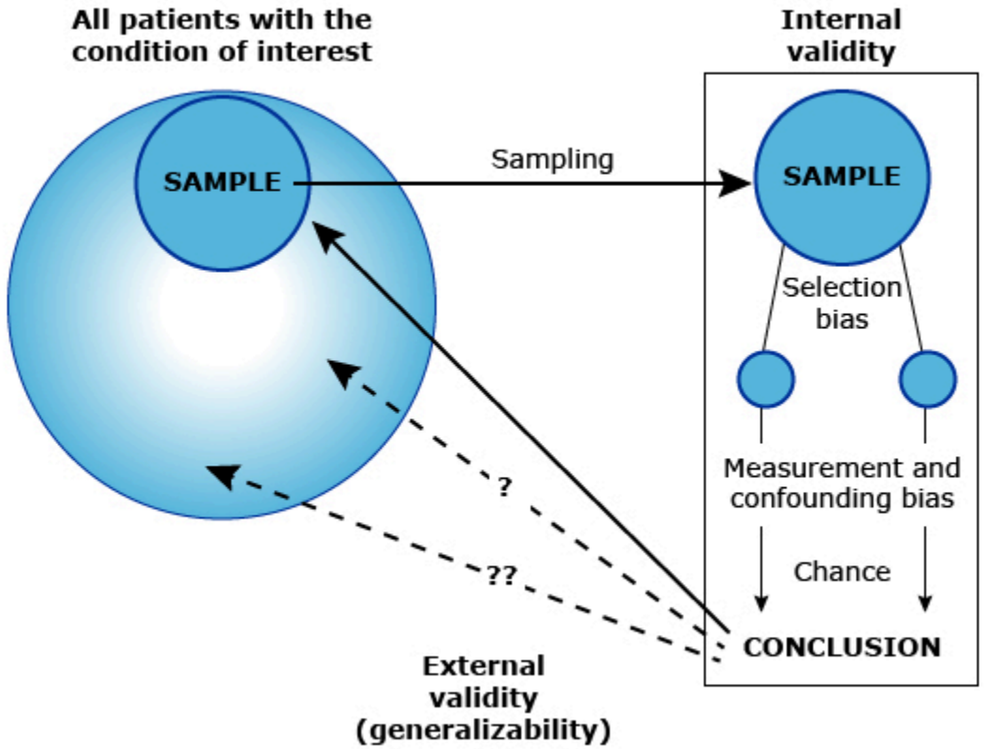
- Review:
  - Ecologic study: dietary fat & breast cancer mortality
  - RCT: low-fat dietary intervention
- Quick question
  - Which design offers stronger evidence for causality?
- Follow-up
  - What might still limit confidence in either study?
- Key idea: Even strong study designs have limitations. That leads us to validity.

Challenges to the Validity of Study Designs

- Even well-designed studies can produce misleading results.
- Two central questions:
  - External validity: Can the findings be generalized to other populations or settings?
  - Internal validity: Is the observed association likely to reflect a true causal relationship?
- If validity is compromised, conclusions may reflect:
  - Random error (chance)
  - Systematic error
  - Confounding
- Key idea: Strong design does not eliminate threats to validity.

External Validity

- Definition: The extent to which study results can be generalized to other populations or settings
- Key influences
  - Sampling error: random variation
  - Sampling bias: systematic lack of representativeness
- Guiding question: Do these results apply beyond this sample?



Sampling Error

- Definition: Random variation that occurs because we study a sample, not the entire population.
- Key points
  - Occurs whenever a sample is drawn
  - Affects the precision of estimates
  - Not a design flaw — a natural result of sampling
- How to reduce it: Increase sample size
- Example
  - If the true population smoking rate is 20%,
  - A sample of 50 people might estimate 16% or 24% due to chance.
- Key takeaway: Sampling error reflects random variation, not systematic bias.

Sampling Bias

- Definition: A systematic error that occurs when the sample is not representative of the target population.
- Key points
  - Results from how participants are selected
  - Affects both generalizability and potentially conclusions
  - Unlike sampling error, it does not disappear with larger sample size
- Example
  - Studying hypertension using patients from a single specialty clinic may overrepresent more severe cases.
- Key takeaway: Sampling bias threatens external validity because the sample does not reflect the population of interest.

Internal Validity

Random Error vs Systematic Error

Random Error (Chance)

Natural variability in data

Directly affects precision (spread of estimates)

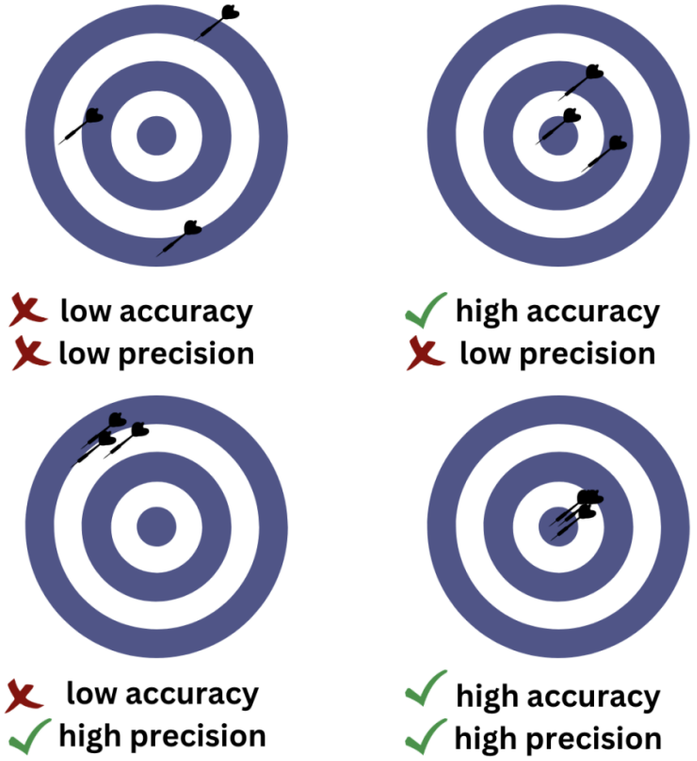
Reduced by increasing sample size

Systematic Error (Bias)

Flaws in design, selection, or measurement

Affects accuracy (closeness to the truth)

Not corrected by larger sample size



Exercise: Random or Systematic?

For each scenario, decide: Random error (chance) or Systematic error (bias)

- A random sample of 40 students is used to estimate smoking prevalence on campus. The study finds 16% prevalence, while the true campus rate is 20%.
- In a survey about alcohol use, participants are asked face-to-face by a researcher. Many participants report lower drinking levels than expected.
- In a cohort study of pesticide exposure and Parkinson's disease, more participants in the exposed group drop out during follow-up than in the unexposed group.
- Blood pressure is measured using two different devices. One device consistently gives readings 5 mmHg higher than the other.

Bias (Systematic Error)

- Definition: A systematic error in the design, conduct, or measurement of a study that distorts the true association.
- Key features
  - Shifts results in a consistent direction
  - Affects accuracy, not precision
  - Not corrected by increasing sample size
- Two major categories
  - Selection bias
  - Information bias
- Key idea: Bias creates a misleading estimate of the exposure–outcome relationship.

Selection Bias

- Definition: A systematic error caused by differences in who enters or remains in the study.
- How it occurs
  - Differences in participant selection
  - Differences in participation or refusal
  - Differences in loss to follow-up
- Key idea: When the groups being compared differ because of the selection or retention process, the association may be distorted.

Common Types of Selection Bias

- Self-selection bias: Participation or refusal is related to exposure or outcome
- Healthy worker effect: Workers tend to be healthier than the general population
- Control selection bias: In case-control studies, controls do not represent the source population
- Loss to follow-up (attrition bias): In cohort studies, dropout differs between exposure groups
- Key idea: Selection bias occurs when group differences arise from who enters or remains in the study.

Information Bias

- Definition
  A systematic error caused by inaccurate or inconsistent measurement of exposure or outcome.
- How it occurs
  - Differences in how information is reported
  - Differences in how information is collected
  - Differences in how information is recorded
- Key idea: When exposure or outcome data differ between groups, the association may be distorted.

Types of Information Bias: Recall Bias

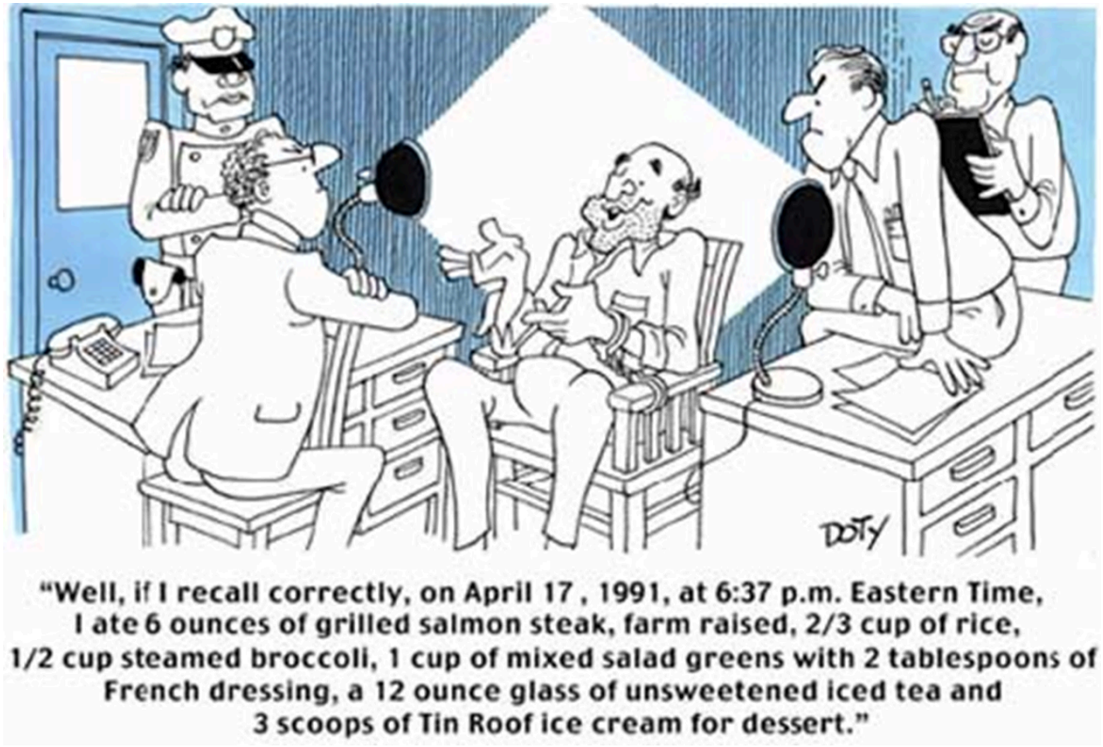
Definition: A type of information bias in which participants remember past exposures differently.

When it occurs

Often in case-control studies

Cases may recall exposures more carefully than controls

Key idea: Differences in memory between groups can distort the exposure–outcome association.



Types of Information Bias: Researcher Bias

Interviewer Bias

Occurs during data collection

Interviewer knowledge influences how questions are asked

Can lead to different responses between groups

Observer Bias

Occurs during outcome assessment

Observer expectations influence how outcomes are recorded

More likely when blinding is absent

Key idea: When researchers influence data collection or measurement, results may be systematically distorted.



Types of Information Bias: Social Desirability

- Definition: participants provide responses that are viewed as socially acceptable rather than accurate.
- When it occurs
  - Sensitive behaviors (e.g., smoking, alcohol use, drug use)
  - Self-reported health behaviors or lifestyle factors
- Key idea: When participants modify responses to appear favorable, exposure or outcome data may be systematically distorted.


QUESTIONNAIRE
DOTY

Types of Information Bias: Hawthorne effect

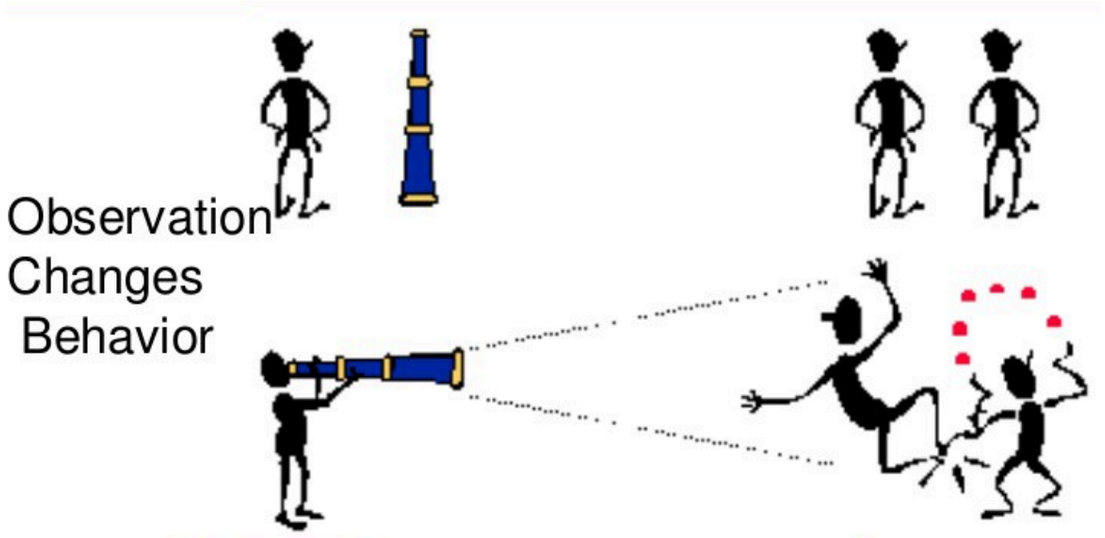
Definition: participants change behavior because they know they are being observed.

When it occurs

During clinical trials or behavioral studies

When participants are monitored or aware of measurement

Key idea: when behavior changes due to observation, the measured exposure or outcome may not reflect usual behavior.



Exercise: Identify the Type of Information Bias

For each scenario, identify the most likely type of information bias and explain why.

- In a case-control study, parents of sick children recall pesticide exposure during pregnancy more carefully than parents of healthy children.
- Participants report lower alcohol use in face-to-face interviews than expected from sales data.
- In a clinical trial, outcome assessors know which participants received the treatment and rate them more favorably.
- Interviewers ask more detailed exposure questions to lung cancer cases than to controls.
- Participants increase exercise levels during a study because they know their activity is being monitored.

Reducing Selection and Information Bias

- Reducing Selection Bias
  - Use appropriate comparison groups
  - Apply clear and consistent inclusion criteria
  - Minimize loss to follow-up
  - Ensure recruitment is not related to both exposure and outcome
- Reducing Information Bias
  - Use standardized definitions and procedures
  - Apply blinding when possible
  - Collect data consistently across groups
  - Use validated measurement tools
- Key idea: Careful design and consistent measurement strengthen internal validity.

Confounding

- Definition: a distortion of the exposure–outcome association caused by a third variable.
- How to identify a confounder
  - Associated with the exposure
  - Independently associated with the outcome
  - Not caused by the exposure
- Example
  - Smoking may confound the association between alcohol use and lung cancer.
- Key idea: If a third variable meets these criteria, the observed association may not reflect a true causal relationship.

Scenario

Studying the effect of Alcohol Use (exposure) on Lung Cancer (outcome).

Confounder: Smoking

Associated with alcohol use

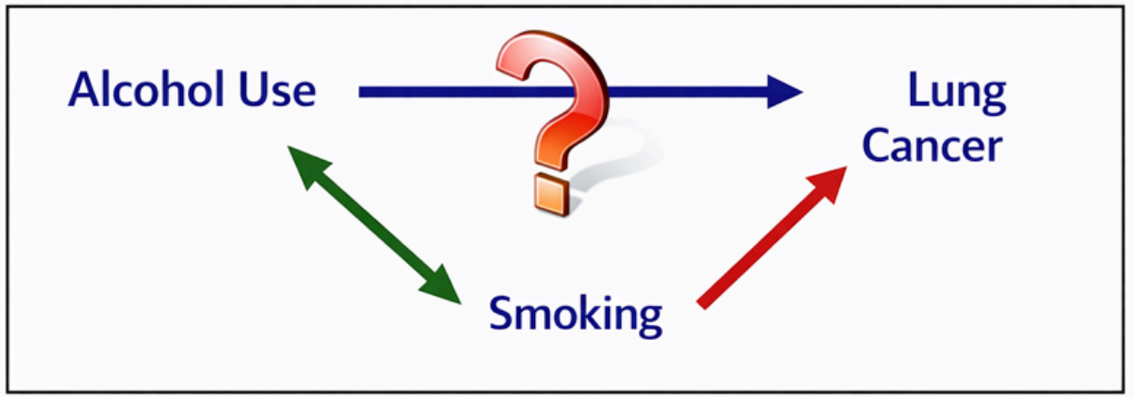
Independently associated with lung cancer

Not caused by alcohol use

Effect

The observed association between alcohol and lung cancer may be distorted because smoking is related to both.

Example of Confounding



- ✓ Associated with the exposure (Alcohol use)
- ✓ Independently associated with the outcome (Lung cancer)
- ✓ Not on the causal pathway

Exercise: Could This Be Confounding?

For each scenario, decide whether the association could be explained by confounding. Identify a possible confounder and explain why.

- Scenario 1: A study finds that people who carry lighters have a higher risk of lung cancer.
  - What third variable might explain this association?
  - Does it meet the criteria for a confounder?
- Scenario 2: A study finds that children who sleep with the light on are more likely to develop myopia (nearsightedness).
  - What third variable might explain this association?
  - Does it meet the criteria for a confounder?

Addressing Confounding

- Confounding cannot be ignored when interpreting results.
  - Randomization helps balance confounders between groups
  - In observational studies, researchers must identify and measure potential confounders
  - Results should be interpreted cautiously if important confounders are not accounted for
- Key idea: If confounding is not addressed, an observed association may not reflect a true causal relationship.

Wrap-Up

Causal inference depends on:

- Study design
  - Observational vs experimental
  - Ecologic vs RCT
- Validity
  - External validity
  - Internal validity

Threats to internal validity

- Chance
- Selection bias
- Information bias
- Confounding

Key message: Strong conclusions require both good design and strong validity.

Final Epi Presentation
Get started!